

Snowflake Developer

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Summary:

- Experienced Snowflake Developer with expertise in modernizing data platforms, migrating legacy systems to cloud-native solutions, and optimizing data workflows for improved performance and scalability. Proven ability to deliver high-impact data migration and analytics projects in industries like mortgage and healthcare.
- Proficient in Data Engineering & ETL/ELT Pipelines, specializing in Snowflake, Informatica Cloud, dbt, Azure Data Factory, and AWS Glue. Skilled at designing, implementing, and optimizing complex data pipelines for large-scale, high-performance data processing.
- Cloud Data Platform Expertise in Snowflake, AWS (S3, Redshift, Lambda, Glue), and Azure (Data Lake Storage, Power BI), with a strong focus on data integration, real-time data ingestion (using Snowpipe and Event Grid), and workload management for cost-effective, scalable solutions.
- Advanced SQL & Python Developer, proficient in writing complex queries, stored procedures, and custom data transformations using Python (boto3, pandas, pyodbc) to automate ETL workflows and enhance business logic in data pipelines.
- Data Security & Compliance Advocate, with hands-on experience implementing Role-Based Access Control (RBAC), data masking, encryption, and adherence to regulatory standards like HIPAA and GDPR to ensure secure and compliant data handling.
- Data Visualization and Reporting Expertise, leveraging Power BI and Tableau to create interactive dashboards and real-time reporting solutions for business stakeholders. Proven ability to integrate data from multiple sources and provide actionable insights for decision-making.
- Skilled in Cloud-Native Data Warehousing using Snowflake features like Time Travel, Zero-Copy Cloning, materialized views, and clustering keys to optimize performance, reduce costs, and ensure data integrity during migration and processing.
- Strong Collaborative & Documentation Skills, effectively working with cross-functional teams, including data analysts, engineers, and business stakeholders, to design scalable data solutions and document processes in Confluence for future reference and knowledge sharing.
- Proven Track Record in Delivering Large-Scale Data Migrations with a focus on data integrity, performance optimization, and zero downtime. Successfully migrated over 10 TB of healthcare data to Snowflake, reducing operational delays and improving query performance by 50%.
- Expert in Data Monitoring & Troubleshooting, using tools like Snowflake Query Profiler, Azure Monitor, and AWS CloudWatch to monitor, analyze, and optimize ETL performance, ensuring fault tolerance, scalability, and efficient data workflows.
- Automation & Orchestration Expertise using AWS Lambda, Azure Data Factory, and Python scripting to automate data processing workflows and enhance operational efficiency.

Technical Skills:

Category	Skills/Tools/Technologies
Data Warehousing & Cloud	Snowflake, Snowpipe, Clustering Keys, Materialized Views, Virtual Warehouses, SnowSQL
Cloud Platforms	Azure Cloud, AWS (S3, Redshift, Lambda, Glue, CloudWatch), Azure Data Lake Storage
ETL & Data Integration	Informatica Cloud, dbt, Azure Data Factory, AWS Glue, Python (boto3, pyodbc, pandas)
Data Security & Compliance	RBAC, Column-Level Encryption, Dynamic Data Masking, HIPAA, GDPR, Azure Key Vault
Data Visualization	Power BI, Tableau, Azure Power BI
Monitoring & Optimization	Query Profiler, Azure Monitor, CloudWatch, Informatica Monitoring
Data Formats	CSV, JSON, Parquet

Automation & Orchestration	AWS Lambda, Python (boto3), pyodbc
Data Modeling & Analytics	Star/Snowflake Schema, Data Marts, Data Transformation
Version Control & Collaboration	Confluence, Jira

Professional Experience:

Client: Mr. Cooper, TX

June 2024 – Till Date

Role: Snowflake Developer

Project: Mortgage Data Lake Modernization with Snowflake

Overview: The project focuses on modernizing the existing Mortgage Data Lake by leveraging Snowflake's cloud-native data warehousing capabilities. It involves migrating legacy data systems to a scalable, high-performance data platform to enhance data accessibility, improve analytics capabilities, and support real-time insights for mortgage operations.

Responsibilities:

- Migrate mortgage data from Azure Data Lake Storage (ADLS) and other cloud-based platforms to Snowflake using Informatica Cloud and dbt for scalable transformations.
- Design and implement star and snowflake schema models in Snowflake, ensuring adherence to best practices for performance and maintainability.
- Leverage Snowflake Time Travel and Zero-Copy Cloning for data backup and version control during the migration process.
- Develop custom SQL scripts and Snowflake stored procedures to support data migration and validation tasks.
- Build and optimize ETL/ELT pipelines using Informatica Cloud, dbt (Data Build Tool), and Azure Data Factory (ADF) for efficient data extraction, transformation, and loading into Snowflake.
- Utilize Snowflake Snowpipe for real-time data ingestion and Azure Event Grid for triggering pipelines on data arrival.
- Write advanced SQL for data transformations and implement Python-based scripts in dbt for custom business logic.
- Ensure pipelines are fault-tolerant, monitorable, and optimized for large-scale data processing.
- Optimize Snowflake query performance by configuring clustering keys, materialized views, and result caching.
- Design and configure virtual warehouses in Snowflake for cost-effective resource scaling and workload management.
- Monitor and analyze pipeline performance using Snowflake Query Profiler, Azure Monitor, and Informatica Monitoring tools.
- Implement Role-Based Access Control (RBAC), column-level encryption, and Dynamic Data Masking in Snowflake to ensure mortgage data security.
- Enforce compliance with regulatory standards (GDPR, CCPA) using Azure Key Vault for secure key management and Snowflake's Data Sharing for controlled access.
- Integrate Snowflake with Azure Power BI for advanced mortgage data reporting and visualization.
- Collaborate with business analysts, data scientists, and cross-functional teams to translate business requirements into scalable Snowflake data solutions.
- Support ad-hoc analytics by creating reusable views and data marts in Snowflake tailored to business needs.
- Document data models, pipeline designs, and ETL/ELT workflows in Confluence and ensure the knowledge base is up to date.
- Troubleshoot pipeline failures and resolve issues using tools like Azure Log Analytics and Snowflake Incident Manager.

Used Tools: Snowflake, Azure Cloud, Azure Data Lake Storage, Informatica Cloud, dbt, Azure Data Factory, SQL, Python, SnowSQL, Snowflake Clustering Keys, Materialized Views, Snowflake Query Profiler, Azure Key Vault, Role-Based Access Control (RBAC), Dynamic Data Masking, Power BI, Azure Monitor, Confluence, Jira.

Client: Gain Well Technologies, Arkansas

March 2023 – May 2024

Role: Snowflake developer

Project: Healthcare Data Analytics Platform Migration

Objective: To modernize the healthcare data platform by migrating the client's legacy systems to Snowflake Data Cloud. The goal was to enable scalable, real-time analytics and enhance data processing efficiency for healthcare claims, patient insights, and provider operations.

Responsibilities:

- Migrated healthcare data from on-premise systems (e.g., Oracle and SQL Server) to Snowflake.
- Integrated data from disparate sources, such as Electronic Health Records (EHRs), Claims Management Systems, and external healthcare databases.
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- Integrated data from disparate sources, such as Electronic Health Records (EHRs), Claims Management Systems, and external healthcare databases.
- Built robust data pipelines using Informatica Cloud and Snowflake native features like Snowpipe for continuous data ingestion.
- Leveraged Python and SQL to create stored procedures for complex data transformation.
- Optimized query performance by implementing clustering keys, result caching, and materialized views.
- Conducted workload management by monitoring Query History and adjusting virtual warehouse sizing.
- Ensured compliance with HIPAA regulations by implementing robust data masking and encryption policies.
- Configured Role-Based Access Control (RBAC) for secure access to sensitive patient and claims data.
- Supported healthcare analysts by preparing datasets for visualization tools like Tableau and Power BI.
- Delivered actionable insights by creating real-time dashboards for monitoring patient claims and operational metrics.
- Worked closely with healthcare domain experts to define business requirements.
- Provided production support for Snowflake environments, resolving data quality and system performance issues.
- Migrated over 10 TB of healthcare data to Snowflake within a 6-month timeline, ensuring zero data loss.
- Improved query performance by 50%, enabling faster analytics for patient claims and billing cycles.
- Implemented a real-time claims processing pipeline that reduced operational delays by 30%.

Used Tools: Snowflake Data Cloud, Informatica Cloud, SnowSQL, Snowpipe, SQL, Python, Tableau, Power BI, AWS S3, RBAC, Data Masking, Encryption, HIPAA, GDPR.

Client: Tricon, India

July 2018 – Aug 2022

Role: ETL Developer

Project: Data Integration using Python on AWS

Overview: Led the development of an automated data integration pipeline using Python and AWS services. The pipeline extracts, transforms, and loads (ETL) data from diverse sources (e.g., databases, APIs, cloud storage) into Amazon Redshift and S3 for reporting and analysis. Optimized and automated workflows to improve data processing efficiency and ensure data accuracy.

Responsibilities:

- Developed Python scripts using boto3, pyodbc, pandas, and requests to extract data from various sources including relational databases, external APIs, and cloud storage Amazon S3.
- Utilized AWS services like boto3 to interact with Amazon S3 and Amazon Redshift for data transfer and integration.
- Performed data cleaning, validation, and transformation using Python libraries such as pandas, numpy, and pySpark to ensure data quality and compliance with business requirements.
- Wrote custom Python functions for complex data transformations, including filtering, aggregation, and data type conversions.
- Loaded transformed data into Amazon Redshift using Python and psycopg2, employing COPY commands for bulk loading and SQL insert queries for smaller datasets.
- Used Python scripts to upload processed data into Amazon S3 in formats like CSV, Parquet, and JSON, enabling further processing or storage.
- Automated ETL processes using AWS Glue, scheduling Python-based ETL jobs to ensure seamless data workflows.

- Leveraged AWS Lambda to trigger real-time data processing and AWS CloudWatch for monitoring job status and performance.
- Implemented error handling and logging mechanisms in Python scripts to detect and troubleshoot issues, ensuring data pipeline reliability.
- Used AWS CloudWatch for centralized logging and monitoring of ETL job executions, setting up alarms for failures.
- Developed Python-based validation checks to ensure data integrity during each stage of the ETL pipeline.
- Created reconciliation scripts to compare data at various stages, identifying and resolving discrepancies.
- Optimized Python ETL scripts for processing large volumes of data using techniques like parallel processing (multiprocessing), efficient data structures, and batching.
- Continuously monitored performance and fine-tuned ETL workflows to improve efficiency.
- Worked closely with data engineers, analysts, and other teams to gather requirements and ensure the ETL solution met business needs.
- Documented Python scripts, ETL workflows, data transformations, and AWS configurations for future maintenance and knowledge sharing.

Used Tools: AWS Glue, Amazon S3, Amazon Redshift, AWS Lambda, AWS CloudWatch, Python, SQL, AWS CLI, psycopg2, AWS SDK for Python (boto3), requests, CSV, JSON, Parquet, SQL, Amazon S3, Amazon Redshift.

Education:

Masters - University of North Texas – USA

Bachelors - Koneru Lakshmaiah University – India
